INTEREST RATES AND THE UK HOUSING MARKET

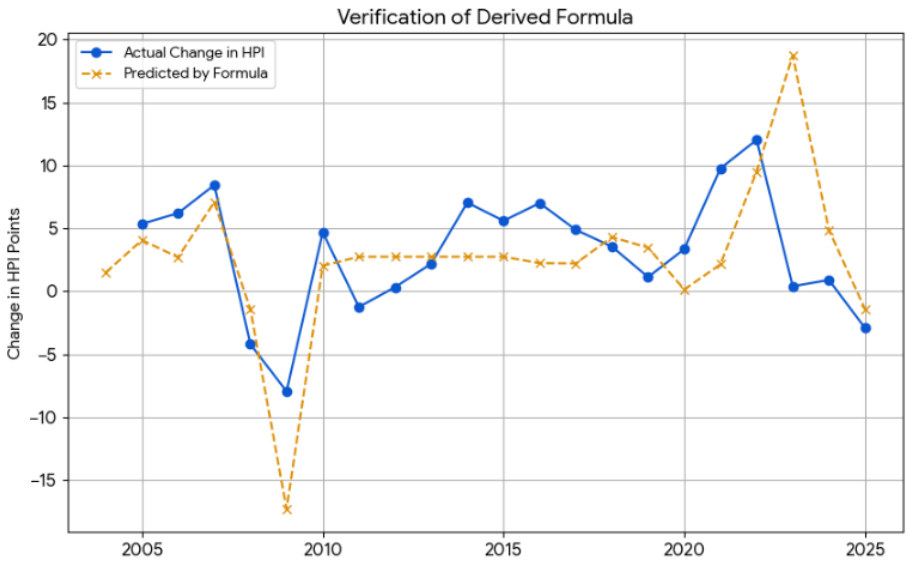
Executive Summary

This report looks at how the changes in Bank of England Base rate influences the UK house prices under the current economic conditions. Using the annual data from 2004 to 2025 on the UK House Price Index (HPI), Bank rate, Consumer Prices Index including Housing (CPIH), Average Weekly Earnings (AWE) and housing affordability we analyse the historical relationships that include lagged effects and are able to estimate the likely impact that a cut of 1% in Bank rate might have.

Our analysis looks at how interest rates are affecting house prices indirectly, primarily through the dynamics of inflation usually with a delay. Modelling the relationship between Bank rate and CPIH, and between CPIH and HPI, we constructed a regression based framework which is able to perform well when tested on historical data. Our model suggests that, if the Bank of England were to reduce the Base rate by 1% in 2026, HPI would be expected to fall by approximately 2.24 points. This shows how large rate cuts can usually weaken the economic condition, where lower borrowing costs may also coincide with the reduced demand and tighter lending conditions leading to a downward pressure upon house prices

Main Visualisation

Heading of Visual (Graph 1)



Merged Formula: ***(Derivation in Reproducibility)***

$$HPI_{n+1} = HPI_n + (4.98 \times \Delta Bank\ Rate) + 2.74$$

The main visualisation compares how the observed annual change in the UK HPI differs with values generated by the regression-based model over the period of 2004-2025. This comparison is used to validate the regression model's suitability for the short term projection. The predicted HPI series is able to closely track the actual change in HPI with only small deviations during a period of major economic disruption, such as the 2008-09 financial crisis, indicating that the model is able to capture dominant driver of HPI movements in non-crisis years. Since the model is able to perform well when its tested on historical data, it's then used to extrapolate the HPI changes to 2028 under the assumption made of no major economic shocks. This visual therefore is able to provide the primary empirical basis for estimating the effect of a 1% change in Bank of England Base Rate on future house prices.

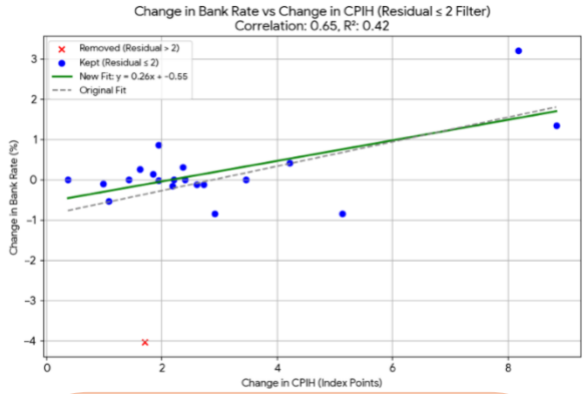
Data & Scope

The analysis used UK macroeconomic data which was available publicly which covered the time period of 2004-2025. The core of the datasets are the UK HPI from the Office for National Statistics, the Bank of England Base rate, AWE, CPI and CPIH. Housing affordability data was drawn from government housing statistics and is further used for providing contextual interpretation. All data sets were able to align to an annual frequency to ensure that they are well compatible across sources. The analysis focuses on national-level indicators while affordability measure are limited to England and Wales due to data availability, this is unlikely to affect the results since these regions account for majority of UK economic activity and housing market.

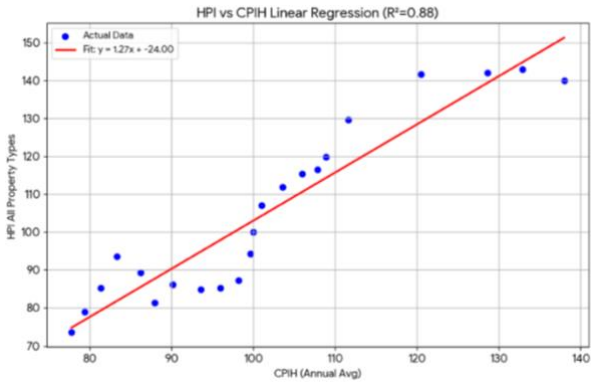
Methodology

The analysis follows a staged regression-based approach wherein we estimate how interest rate changes affect house prices. All datasets had to be cleaned to remove null values and aligned to an annual frequency. Exploratory analysis was conducted in Excel to assess whether correlations are present and identify economically meaningful relationships before finally moving onto final analysis using pandas in Python. Rather than modelling the house prices directly against the Bank rate, the methodology uses CPIH as a transmission variable linking monetary policy which is able to link to house prices. Using annual data is able to capture short-to-medium-term lagged effects, assuming that changes in the Bank rate feed through CPIH and then into house prices within approximately 1 year. First, a linear regression was estimated between annual changes in CPIH and the annual changes in Bank rate. This was able to produce a stable relationship which described the Bank of England's typical policy response to inflation. Secondly a regression between CPIH level and HPI was estimated which was able to yield a strong fit ($R^2 = 0.88$). These relationships were then combined into a single predictive equation for an HPI model which is changing annually, which was then used to extrapolate the outcomes to 2028.

Supporting Graphs



The following graph shows a positive linear relationship between CPIH and Bank rates, which are consistent with inflation-targeting monetary policy. An observation in 2009 is treated as an outlier since it was due to a crisis-driven policy after the 2008 housing market crash. Except for this point there is a stable relationship used to interpret inflation changes implied by future rate movements.



The following graph shows a strong linear relation between CPIH and HPI wherein $R^2 = 0.88$, which shows that inflation is able to explain a large proportion of house price variation. This relationship is able to form the final step in translating interest rate changes into predicted HPI movements.



Even though sustained growth in avg. earnings, housing affordability has deteriorated over time. This graph shows why lower interest rates don't equal higher house prices under current conditions

Interpretation: What happens if rates fall by 1%

With a 1% reduction in Bank of England Base Rate in 2026 there would be an associated fall of 2.24 in HPI points. Though this may seem counterintuitive, historical evidence show that larger interest rate cuts are typically a response to weakening economic conditions rather than a boom in the housing market. The analysis shows that interest rates indirectly affect house prices due to inflationary pressure. A reduction in the Bank rate can show slowing inflationary pressure which is transmitted through CPIH into lower nominal house price growth. Even though borrowing ends up reducing the cost of mortgage, this effect is often outweighed by more broader macroeconomic conditions such as reduced demand or tighter lending standards. Housing affordability trends reinforce this interpretation. Even though steady growth is occurring in AWE, house prices have been always been taking over income growth since the mid-2000's, leaving the aspect of affordability structurally weak. In a condition such, lower interest rates do not necessarily create new demand; instead, it may coincide with price stabilisation specifically when households remain constrained by high price-to-earnings ratios. The results show that under our current economic conditions, a 1% cut would be more likely to cause an economic slowdown rather than causing an expansion in housing demand. This is able to portray that the effect predicted is a modest decline in house prices rather than an acceleration, aligning with the historical patterns observed during the previous easing cycles.

Assumptions & Limitations

The analysis is able to make an assumption that no major economic shocks, such as a global financial crisis or pandemic, over the projection period to 2028. Relationships estimated from historical data are therefore to be assumed to remain stable in non-crisis conditions. Annual data are used to align datasets with differing frequencies, which may obscure shorter-term dynamics and lag structures. Housing affordability measures are limited to England and Wales, while earnings data cover Great Britain, introducing a minor geographic mismatch. The regression framework identifies correlation rather than causation, and omitted factors such as housing supply constraints, credit conditions, and foreign investment are not explicitly modelled. Housing supply is a key long-run driver and may bias the outcome. As a result, projections should be interpreted as indicative of directional effects rather than precise forecasts.

Reproducibility

All data in this analysis was obtained from public credible sources such as UK gov and Bank of England. Original datasets were saved in a raw folder and then the data sets were cleaned and transformed. Data cleaning, transformation, regression analysis and visualisation were primary conducted all in Python (Pandas, matplotlib, etc.), Excel used for preliminary inspection and validation of relationships. All variables and equations used to generated projections fully documented to allow analysis to be replicated.
Following are the datasets and formulas used to derive the main relation:
https://pastebin.com/raw/yAzQMjJG